

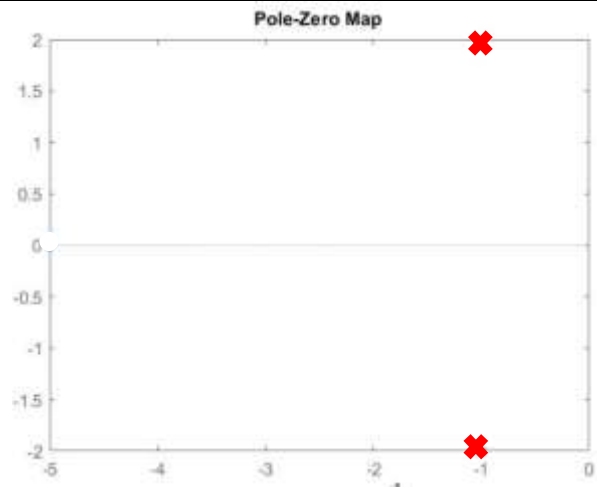
05.06.2020

Time : 10 am- 12 noon

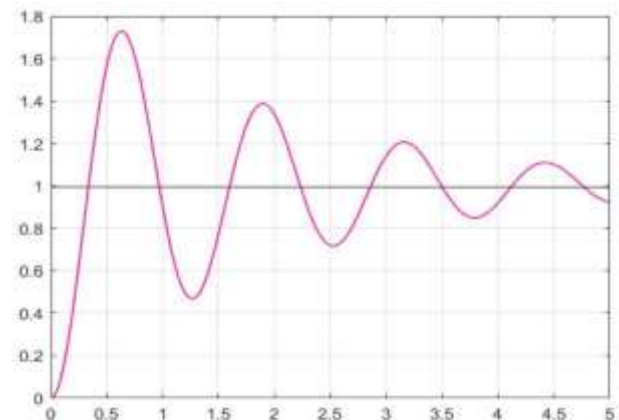
Max. Marks: 50

Write the answer and upload the image of your answer script to iceemscusat@gmail.com before 12.10 pm. You can refer to any material available with you. But, if same mistakes are seen in more than one answer scripts, zero mark will be awarded to all such answer scripts.

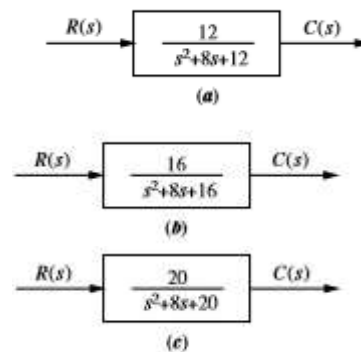
1. The pole-zero map of a system is shown. Determine the following
- The damping factor of the system
 - The natural frequency
 - The transfer function of the system
 - Write down the expression for the natural response of the system and plot (only qualitatively) if the initial conditions are $x(0) = 2 \times 10^{-2}m$, $\dot{x}(0) = 1m/s$



2. The unit step response of a second order system is shown in the figure. Determine the 2% settling time. What is the possible transfer function of the system?
- If you want to reduce the overshoot to 10%, what should be the damping factor? How it affects the rise time, peak time and 2% settling time?



3. For each of the system shown, calculate the damping ratio. Identify what kind of impulse responses are expected of each system and plot them in the same figure (only qualitatively)



4. Determine the rise time, peak time, maximum overshoot, and settling time of the system

$$G(s) = \frac{100}{s^2 + 15s + 100}$$

5. Derive the expression for unit step response of a system whose transfer function is given. Plot the response qualitatively

$$G(s) = \frac{1-2s}{1+s}$$

